

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Subject Code & Name: Hydraulics II (BTCVC401) Max Marks: 60	Branch : Civil Semester :IV Date:	Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.			
A)	List and explain the types of flows in Open Channel?	CO1	6	
B)	Explain and draw typical patterns of velocity distribution in various channel sections.	CO1	6	
C)	Define the terms; 1.Weir 2.Crest 3.Notch	CO1	6	
Q.2	Solve Any Two of the following.			
A)	Determine the most efficient section of a trapezoidal channel with side slopes 1V:2H, carrying a discharge of 11.25 m ³ /s with a velocity of 0.75 m/s. What should be the bed slope of the channel? Take Manning's n =0.025	CO2	6	
B)	Define specific energy and explain in detail with neat sketch of specific energy curve?	CO2	6	
C)	State the conditions under which the rectangular section of an open channel will be most economical. Derive these conditions.	CO2	6	
Q. 3	Solve Any One of the following.			
A)	Derive differential equation of GVF in different forms?	CO3	6	
B)	A hydraulic jump takes place in a horizontal rectangular channel with sequent depths of 0.25 m and 1.50 m at the beginning and end of the jump respectively. Estimate the (i) Discharge per unit width of the channel and (ii) Energy loss.	CO3	6	
C)	Derive an expression of force exerted by jet striking a curved plate?	CO4	6	
Q.4	Solve Any Two of the following.			
A)	A jet of water, 40 mm diameter, strikes a hinged vertical plate of 800 N	CO4	6	

	weight, normally at its centre at 15 m/s. Find i) angle of deflection; ii) force F that must be applied at lower edge to keep the plate vertical.		
B)	Draw neat sketch of Hydroelectric power plant and explain its working?	CO5	6
C)	Draw neat sketch of Francis turbine and explain its working?	CO5	6
Q. 5	Solve Any One of the following.		
A)	What are the types of reciprocating pumps and explain in detail with sketch?	CO5	6
B)	Obtain an expression of the boundary shear stress in terms of momentum thickness.	CO5	6
C)	What do you mean by boundary layer separation? What is the effect of pressure gradient on boundary layer separation?	CO5	6
	*** End ***		

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